Relief load basis

PSV, TSV and vent register of the capture unit

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Teaching document. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data. No vendor, client or contractor data. Checker and approver are left blank on purpose.

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1 Purpose, scope and basis

This relief load basis lists every pressure-relief device the MEA CO2 capture unit needs. For each device it gives the credible overpressure scenarios (API 521), the relief load, the governing case and the required orifice (API 520 Part I, API 526 letters). It also sets the PSV set pressures against the maximum allowable working pressure (MAWP), defines where each device discharges, and records why vapour depressuring is not required. It resolves the HOLD of note 5 on CC-PID-101 rev A.

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1.1 References

- CC-PID-101 rev A (P&ID, note 5: PSVs HOLD)
- MEA-PFD-001 rev 0 (PFD)
- MEA-SA-HZP-001 rev A (HAZOP report)
- MEA-IN-SRS-001 rev 0 (SRS of the SIFs)
- MEA-IN-CE-001 rev 0 (C&E matrix)
- MEA-IN-INL-001 rev 0 (instrument list)
- MEA-ME-EQL-001 rev 0 (equipment list: design pressures used as MAWP)
- API Standard 521 (pressure-relieving and depressuring systems)
- API Standard 520 Part I (sizing of pressure-relief devices)
- API Standard 526 (flanged steel relief valves: effective orifice areas D to T)

1.2 Method

A scenario is credible when a single failure causes it (single jeopardy). No credit is taken for a control loop that would help, or for the SIS. The blocked-outlet family of cases depends on how the plant behaves dynamically, so it is computed with the simulator model itself: each case starts from a copy of the design state (noise off), every trip constant is moved out of reach and the trip function is stubbed (both restored afterwards), and a virtual relief valve holds the reflux drum at the relieving pressure. The relief rate is the outflow that keeps the drum pressure balance of the model at P_1:

$$n_{rel} = (P_d - P_1) \cdot V_{gas} / (R\,T_c) / \Delta t, \quad V_{gas} = 40\ m^3,\ \Delta t = 0.25\ s,\ P_1 = set + 10\ \% = 374.9\ kPa\ a$$

The load is the largest 60-s mean of n_{rel}, given wet (overhead H2O/CO2 ratio, as it leaves the T-102 overhead line where PSV-301 sits) and dry (saturated at the drum temperature, as it would leave V-101). Fire, tube rupture and thermal expansion are equations (API 521); the model has none of them.

1.3 Design point used (model, noise off)

Item	Value	Source
Stripper / reflux drum pressure	175.0 kPa a (PIC-301 SP 175.0)	snapDesign
Reboiler temperature	120.0 °C	snapDesign
LP steam supply	350 kPa a, 138.9 °C	K.Ps, K.Ts (assumed in the OTS)
CO2 product FI-304	12.50 t/h	snapDesign
Lean solvent FIC-201	269.7 t/h	snapDesign
PSHH-301 trip (I-2)	250 kPa a	K.Pshh
PIC-301 high alarm / LIC-208 H / LIC-303 H	190 kPa a / 70 % / 75 %	ALM list

2 Device register

One row per device the unit needs, including the devices this basis finds are not required, with the reason. MAWPs are the design pressures of the equipment list MEA-ME-EQL-001; the MP re-rating of the E-102 steam side is this basis's proposal (assumed).

Device	Location (CC-PID-101 line)	Protects	Status	Set kPa a	MAWP kPa a	Governing case	Relief load	Area mm²	Orifice	Discharge to
PSV-301	T-102 overhead line DN500-OV-301, upstream of E-103	T-102, E-102 shell (process side), E-103 process side, V-101	Required	350	371.3 (EQL ME A-ME-EQL-001)	R1 Blocked outlet: PV-301 closed; FV-207 held at its normal opening	8,848 kg/h	3,096	P (4,116 mm²)	Atmosphere through the CO2 vent stack VS-301 at a high point (see part 6)
PSV-302	V-101 vapour space	V-101	Not required (covered by PSV-301) V-101 is open to T-102 through E-103 with no block valve on CC-PID-101; PSV-301 covers it and its fire load is in the PSV-301 fire case. Required only if a block valve is added between T-102 and V-101 (EQ-R05).	—	—	—	—	—	—	—
PSV-207	E-102 tube side: LP steam line DN400-LS-207 downstream of FV-207 / XV-207	E-102 steam side, condensate pot	Required with the EQL rating; not required if the steam side is re-rated for MP (recommended) The EQL rates the E-102 tube side for 420 kPa g, which a letdown failure (MP steam in the LP header) exceeds. Either fit PSV-207 sized for FV-207 wide open on MP steam (large, part 4) or re-rate the steam side and the FV-207 downstream piping for the MP header (recommended, EQ-R06).	521.3 if fitted	521.3 (EQL ME A-ME-EQL-001) ; 1,101 if re-rated (assumed)	LP letdown failure, FV-207 wide open on MP steam	EQL rating: 134,286 kg/h · re-rated: none	46,576	3 × T	Atmosphere at a safe height (steam) if fitted
TSV-501	E-101 rich (cold) side, between the inlet and outlet block valves	E-101 rich side	Required	681	681.3 (EQL ME A-ME-EQL-001)	Thermal expansion, Φ 18.45 MW	138.2 L/min	104.6	E	Closed amine drain / back to the T-101 sump (not to grade)
TSV-502	E-104 cooling-water side	E-104 CW side	Required if the CW side can be blocked in	571	571.3 (EQL ME A-ME-EQL-001)	Thermal expansion, Φ 3.13 MW	15.8 L/min	12.6	D	CW return / open drain
TSV-503	E-103 cooling-water side	E-103 CW side	Required if the CW side can be blocked in	571	571.3 (EQL ME A-ME-EQL-001)	Thermal expansion, Φ 4.08 MW	20.6 L/min	16.4	D	CW return / open drain
TSV-504	E-105 cooling-water side	E-105 CW side	Required if the CW side can be blocked in	571	571.3 (EQL ME A-ME-EQL-001)	Thermal expansion, Φ 5.77 MW	29.2 L/min	23.2	D	CW return / open drain
T-101 (no device)	Absorber T-101	T-101	No PSV; design for full vacuum T-101 is open to the stack, so a PSV has nothing to relieve; the most it can see is the blower shut-off pressure, inside the EQL design pressure. The real hazard is vacuum: steam-out followed by cold solvent condenses the steam faster than any practical vacuum breaker can admit air.	—	—	Blower shut-off; vacuum on steam-out	—	—	—	Treated-gas outlet open to the stack
HV-305 / VS-301	CO2 product line DN300-CO-304 upstream of compression; vent stack VS-301	Start-up and compressor-trip venting of the CO2 product	Required (new)	—	—	CO2 compression unavailable: full product	12,728 kg/h	—	—	VS-301 at a high point, vertical, shared with the PSV-301 tail pipe

3 PSV-301: credible scenarios and the governing case

PSV-301 on the T-102 overhead line DN500-OV-301 protects T-102, the E-102 shell (the process side of the kettle, open to the column), the process side of E-103 and V-101: nothing can isolate any of them from the others. Model cases R1–R7 ran 20 simulated minutes from the design state with all trips disabled.

Case	Scenario	API 521 basis	Cred.	Lifts at	Peak / P1 kPa a	Load wet (dry) kg/h	T °C	M	k	Area mm²	P passes kg/h
R1 governing	Blocked outlet: PV-301 closed; FV-207 held at its normal opening	Blocked outlet (API 521)	yes	170 s	374.9	8,848 (7,147)	114.7	34.2	1.299	3,096	11,763 ✓
R2	Blocked outlet: PV-301 closed; TIC-206 → FIC-207 cascade left in AUTO	Blocked outlet with credit for the control response (comparison only)	no	222 s	374.9	4,546 (3,831)	109.4	35.8	1.297	1,545	—
R3	Blocked outlet after the I-2 trip: PV-301 closed and XV-207 shut at t = 0	Blocked outlet, steam already isolated (rich feed flashing only)	yes	481 s	374.9	260 (240)	92.5	39.5	1.292	82	12,996 ✓
R4	FV-207 fails wide open (FIC-207 MAN 100 %); PV-301 held at its normal opening	Control valve failure (API 521)	yes	—	198.1 (no lift)	0	—	—	—	—	—
R5	FV-207 wide open AND PV-301 closed (two failures, bounding only)	Double jeopardy, not a design case (comparison only)	no	165 s	374.9	8,847 (7,147)	114.7	34.2	1.299	3,096	—
R6	Loss of cooling water to E-103 (TV-302 closed); steam and PV-301 held at normal	Loss of cooling (API 521)	yes	—	262.7 (no lift)	0	—	—	—	—	—
R7	General power failure: all motors and the CW pumps stop; steam stays on, PV-301 held	Utility failure, electric power (API 521)	yes	316 s	374.9	2,076 (1,693)	136.1	21.2	1.318	942	9,066 ✓
F1	External fire on T-102, the E-102 shell and V-101 (wetted area 57.5 m²)	External fire (API 521), 21 % overpressure	yes	API 521 equation	402.2	2,053	144.1	21.4	1.318	874	9,667 ✓
TR	E-102 tube rupture at the normal LP steam pressure (350 kPa a)	Tube rupture (API 521)	yes	orifice equation	— (no lift)	0	—	—	—	—	—
TR2	E-103 tube rupture: cooling water at 400 kPa a into the condensing side	Tube rupture (API 521), liquid displacing vapour	yes	orifice equation	374.9	39	40.0	43.5	1.287	11	14,715 ✓

R2 (credit for TIC-206) and R5 (two failures) are not design cases. R5 gives the same load as R1 (8,847 vs 8,848 kg/h): at the relieving pressure the solvent boils at 133.1 °C, close to the 138.9 °C of the LP steam, so the reboiler duty is set by UA·ΔT (8.16 MW against 13.69 MW at design), not by the opening of FV-207. R4 and R6 do not lift because PV-301 at its normal opening passes the extra vapour. R3 shows the flashing hot rich feed still lifting PSV-301 after XV-207 has shut (EQ-R04).

3.1 Worked calculation · R1 (API 520 Part I, critical vapour flow)

P_2/P_1 = 101.3 / 374.9 = 0.270 < critical ratio $(2/(k+1))^{k/(k-1)}$ = 0.546 → critical flow; C = 0.03948 √(k $(2/(k+1))^{(k+1)/(k-1)}$) = 0.02634 (k = 1.299)

A = W / (C K_d P_1 K_b K_c) · √(T Z / M) = 8,848 / (0.02634 × 0.975 × 374.9 × 1 × 1) × √(387.9 × 1 / 34.19) = **3,096 mm²**

Next API 526 orifice: **P** = 4,116 mm² (75 % used). Dry basis (V-101 outlet, 40.0 °C, M 43.5): 1,999 mm².

3.2 External fire (API 521)

Wetted surface within 7.6 m of grade	Basis	Area m²
T-102 sump + E-102 shell (the model lumps them)	D 3.4 m (K.Dr), liquid to LIC-208 H 70 % of 2.5 m → 1.75 m; side + bottom	27.8
T-102 packed shell (liquid film)	D 2.6 m, 1.6 m inside the zone (packing from 6 m, assumed)	13.1
V-101, horizontal at grade (assumed)	D 1.6 × L 4 m (8.0 m³), liquid to LIC-303 H 75 %	16.6
Total		**57.5**

Q = 43 200 F $A_{ws}^{0.82}$ = 43 200 × 1 × $57.5^{0.82}$ = **1.198 MW** (adequate drainage and fire fighting). At P_1 = set + 21 % = 402.2 kPa a the lean solvent (α 0.246) boils at 144.1 °C with y_{CO2} = 0.129; V = Q / (y ΔH_{abs} + (1 − y) λ) = 0.0267 kmol/s, W = **2,053 kg/h** (M 21.4), A = **874 mm²**, so fire does not govern. The fire relieving pressure is above the

model's 400 kPa a clamp, which is why this case is an equation (EQ-R03).

3.3 Tube rupture (E-102, E-103, E-104)

With the EQL design pressures the 10/13 rule (API 521) leaves three ruptures credible. **E-102**: shell / tubes = 0.643; at the normal LP pressure (350 kPa a) the steam cannot enter a shell relieving at 374.9 kPa a: 0 kg/h. With MP steam in the header at the same time (double jeopardy), two open tube ends (ID 15.75 mm, K_d 0.62, assumed) would pass 1,279 kg/h, well inside the selected orifice. **E-103**: shell / CW = 0.574; cooling water at 400 kPa a (EQL) enters across 25.1 kPa: 6,148 kg/h of liquid that displaces 39 kg/h of vapour (row TR2) and fills V-101. **E-104**: CW / lean = 0.610, but the lean side's maximum operating pressure (495 kPa a) is below the CW-side MAWP (571 kPa a): no overpressure, only amine in the CW return (EQ-R12). No tube rupture governs.

4 Thermal relief, the E-102 steam side and T-101

4.1 Thermal relief valves (API 521 thermal expansion, API 520 liquid sizing)

$q = \alpha_v \Phi / (\rho c)$; $A = 11.78\ Q / (K_d K_w K_c K_v) \sqrt{(G / \Delta P)}$, Q in L/min, ΔP in kPa, K_d 0.65 (certified liquid valve), relieving at set + 10 %, discharge at 0 kPa g. The heat input is the full design duty of the exchanger from the model (conservative).

Device	Blocked-in side	Φ (model duty) kW	ρ kg/m³	c J/(kg K)	μ cP	α_v 1/K (assumed)	Set kPa a	Q L/min	K_v	Area mm²	Orifice
TSV-501	E-101 rich (cold) side, blocked in while the hot lean side flows	18,445	1,112	3,600	2.58	5.00e-04	681	138.2	1.000	104.6	E
TSV-502	E-104 cooling-water side, blocked in with hot lean on the other side	3,131	994	4,180	0.72	3.50e-04	571	15.8	1.000	12.6	D
TSV-503	E-103 cooling-water side, blocked in with overhead vapour condensing	4,079	994	4,180	0.72	3.50e-04	571	20.6	1.000	16.4	D
TSV-504	E-105 cooling-water side, blocked in with wash water on the other side	5,770	994	4,180	0.72	3.50e-04	571	29.2	1.000	23.2	D

4.2 PSV-207 on the E-102 steam side

MP steam 10 bar a / 179.9 °C in the LP header (letdown failure): the demand case of SIF-3 in the SRS tab (assumed there). The model's steam valve law scales its capacity with the supply pressure, so FV-207 wide open would pass 47 × 1,000 / 350 = **134.3 t/h**. The EQL rates the tube side for 521.3 kPa a, below the MP pressure, so with that rating PSV-207 **is required**: P_1 = 563.3 kPa a, $A = 190.5\ W / (P_1 K_d K_b K_c K_N K_{SH})$ = **46,576 mm² = 3 × T** orifices. Re-rating the steam side for 1,101 kPa a (assumed) needs **no PSV**; this basis recommends it (EQ-R06).

4.3 T-101: no PSV; design for full vacuum

The treated gas leaves to the stack with no valve. With the outlet blocked, K-101 at 100 % speed reaches its shut-off head H_0 = 16.2 kPa (model blower curve), so T-101 sees at most 118.5 kPa a, inside its EQL design pressure of 291.3 kPa a: no PSV. Vacuum is the real case: after steam-out, starting the lean pump (269.7 t/h at 40.0 °C) condenses steam at $m\ c_p\ (100 - T_L) / \lambda$ = 7.15 kg/s, which needs 9.9 m³/s of air at 30 °C to replace it. A vacuum breaker cannot admit that: specify T-101 for full vacuum and keep a lock-open vent during steam-out.

4.4 Other scenarios considered

Device	Scenario	Finding
PSV-302	All PSV-301 cases	Covered by PSV-301 while no block valve separates V-101 from T-102.
PSV-207	LP letdown failure: MP steam in the LP header	Credible single failure. With the EQL rating (420 kPa g) the relief load is FV-207 wide open on MP steam; re-rated for MP, no relief.

Device	Scenario	Finding
PSV-207	E-102 tube rupture (steam side is the high side)	Relieves into the shell, so it is a PSV-301 case (part 3).
TSV-501	Blocked-in rich side, hot lean flowing	Thermal expansion (API 521); the heat input is taken as the full design duty (conservative).
TSV-501	Lean (hot) side blocked in	The trapped hot liquid cools down, so its pressure falls. No relief needed.
TSV-501	Plate leak rich → lean	Both sides are liquid at similar pressures, so a leak is a quality problem (fault 10), not an overpressure case.
PSV-301	E-103 tube rupture: cooling water into the condensing side	Credible (EQL: CW side 470 kPa g vs process 270 kPa g, below 10/13). CW at 400 kPa a enters a shell relieving at set + 10 %; the liquid only displaces vapour (part 3): small load, V-101 level rises (LIC-303 H).
TSV-502 / CW return	E-104 tube rupture: lean amine into the CW side	Credible by the 10/13 rule (EQL: CW 470 vs lean 770 kPa g), but the lean side's maximum operating pressure (495 kPa a, P-102 discharge) is below the CW-side MAWP (571 kPa a): no overpressure. The hazard is amine in the CW return (CW-return pH / conductivity detection, TBC).
TSV-502…504	CW side blocked in with the hot side flowing	Thermal expansion; not needed if the CW return block valve is locked open (procedure, typical practice).
T-101	Blocked treated-gas outlet with K-101 at full speed	The pressure is limited to the blower shut-off head, so no PSV is needed: design T-101 for that pressure.
T-101	Steam-out, then cold solvent admitted (vacuum)	Vacuum. The condensation rate makes a vacuum breaker impractical, so specify T-101 for full vacuum and keep a lock-open vent during steam-out.
P-101/102/103/104	Pump dead-head	Centrifugal pumps; shut-off head below the piping design pressure (assumed, TBC on the pump datasheets). No relief.
HV-305 / VS-301	CO2 compression trips or is not yet running	Full CO2 product to atmosphere at height. Dense gas, so the discharge goes vertical and well above platforms.

5 Set pressure vs MAWP

Level	kPa a	kPa g
Normal (PIC-301 SP)	175.0	73.7
PAH (ALM PIC-301 H)	190.0	88.7
PSHH-301 trip (K.Pshh)	250.0	148.7
PSV-301 set	350.0	248.7
MAWP (EQL MEA-ME-EQL-001)	371.3	270.0
Relieving, set + 10 %	374.9	273.5
Model clamp on the drum pressure	400.0	298.7
Fire relieving, set + 21 %	402.2	300.9

PSV-301 is set at 350 kPa a, below the EQL MAWP of 371.3 kPa a. The relieving pressures (set + 10 % / + 21 %) stay inside 110 % / 121 % of the MAWP (398.3 / 428.0 kPa a). The normal pressure is 30 % of the set (gauge), under the 90 % that is typical for a spring valve. The alarm and trip sit between normal and set, so the operator and the SIS act before the valve lifts.

Equipment / side	MAWP kPa a	Source and basis
T-102, E-102 shell (process side), E-103 process side and V-101	371.3	EQL MEA-ME-EQL-001: 270 kPa g + FV (EQL MEA-ME-EQL-001; op + 170 kPa governs)
T-101 shell	291.3	EQL MEA-ME-EQL-001: 190 kPa g (EQL MEA-ME-EQL-001; op + 170 kPa)
E-101 rich side (TSV-501)	681.3	EQL MEA-ME-EQL-001: 580 kPa g (EQL MEA-ME-EQL-001; P-101 shut-off (677 kPa a))
Cooling-water sides of E-103, E-104, E-105	571.3	EQL MEA-ME-EQL-001: 470 kPa g (EQL MEA-ME-EQL-001; CW supply 400 kPa a assumed there)
E-102 tube side (LP steam) as issued	521.3	EQL MEA-ME-EQL-001: 420 kPa g (EQL MEA-ME-EQL-001; LP header 350 kPa a)
E-102 tube side (LP steam), re-rated for the MP header	1,101.3	assumed (re-rating proposed by this RLB): 10 bar g (assumed, proposed by this RLB)

6 Discharge to a safe location and the CO2 vent

- PSV-301 and HV-305 discharge to atmosphere through one vertical vent stack VS-301, at a high point well above any platform within the dispersion distance (typical practice per API 521: a vertical outlet, at least 3 m above platforms nearby). The fluid is CO2 with water vapour; nothing flammable or toxic at the MEA vapour level of the overhead.
- CO2 is about 1.5 times as dense as air. A low release would slump and collect in pits, bunds, trenches and the analyzer house (Consequence tab, CO2 line release). A high, vertical, fast jet mixes before it can come down.
- Solid CO2 (dry ice) cannot form in the tail pipe, because the relieving pressures are below the CO2 triple point (517.95 kPa a), so the expansion goes straight to gas.
- Water condenses in the tail pipe. Fit a drain hole or a low-point drain, a rain cap that does not stop vertical flow, and keep the tail pipe free-draining back to the stack base.
- Thermal relief valves on amine discharge to the closed amine drain, never to grade (MEA, hot loaded solvent). On cooling water they can go to the CW return.

Vent stack VS-301, ID 202.7 mm (assumed): HV-305 full product 12,728 kg/h (M 42.9, 40.0 °C) leaves at **66 m/s**; PSV-301 at the governing load (8,848 kg/h) at **71 m/s**. Relieving pressures 374.9 / 402.2 kPa a are below the CO2 triple point (517.95 kPa a), so no solid CO2 forms.

7 Depressuring

Depressuring (API 521 vapour depressuring) is not required for this unit, for four reasons:

1. The pressures are low: the highest relieving pressure is the fire case at a little over 400 kPa a. Depressuring is aimed at vessels whose wall would be overstressed in a fire before the PSV could protect them, typically at much higher pressures (API 521 typical practice).

2. The inventory is not flammable: CO2, water and 30 wt% aqueous MEA. There is no light hydrocarbon to feed a jet fire, so there is nothing to remove quickly.
3. The fire case is covered by PSV-301 at 21 % overpressure, and the liquid inventory keeps the wetted wall cool while it boils.
4. The SIS removes the main energy source: I-2 shuts XV-207, and the LOPA action SIF-2R (proposed) would also stop P-101 and the flashing rich feed. Venting the stripper quickly would only send hot wet CO2 to atmosphere.

8 Engineering queries

EQ	Subject	Documents that disagree	What	Suggested resolution
EQ-R01	P&ID note 5: "PSV ... E-102 SHELL"	CC-PID-101 note 5; Safety layers (PSV-207); this RLB	In a kettle reboiler the shell is the boiling process side, which is open to T-102, so PSV-301 covers it. The steam is in the tube bundle; whether that side needs a PSV is a different question (PSV-207, EQ-R06).	Revise note 5: "PSV-301 (T-102 overhead) protects T-102, E-102 shell (process side), E-103 and V-101; E-102 tube side see PSV-207 decision; TSV-501...504 thermal".
EQ-R02	PSV-301 set vs MAWP (now agree)	HMI text of PSHH-301 (PSV-301 at 350 kPa a); EQL MEA-ME-EQL-001 (T-102 system 270 kPa g = 371.3 kPa a); this RLB	The EQL now issues the T-102 system design pressure. The set stays at 350 kPa a, below the MAWP, as every other tab uses. Relieving at set + 10 % and fire at set + 21 % stay inside 110 % / 121 % of the MAWP (398.3 / 428.0 kPa a).	Agreed. Keep the set at 350 kPa a and show PSV-301 on CC-PID-101 with the EQL design pressure. This closes the open part of EQL EQ-M03 (blocked outlet within 110 % of the MAWP once PSV-301 is sized for R1).
EQ-R03	Model pressure clamp	src/model.js (P.Pd clamped to 95–400 kPa a); this RLB (fire relieving 402 kPa a)	The model cannot represent the drum above 400 kPa a, so the fire case (21 % overpressure) is computed by the API 521 equation, not by a model run.	Keep the clamp: other tabs store runs that depend on it. The fire case stays an API 521 hand calculation in this RLB.
EQ-R04	I-2 steam trip does not end a blocked outlet	SRS (SIF-2 gap); LOPA (SIF-2R); this RLB run R3	With PV-301 closed and XV-207 already shut, the hot rich feed still flashes and lifts PSV-301 (run R3). The relief device is the last layer for this case.	Keep PSV-301 sized for R1. Close the LOPA action SIF-2R (stop P-101 on PSHH-301), which removes the flashing feed.
EQ-R05	Isolation of V-101 from T-102	CC-PID-101 (no block valve drawn); this RLB (PSV-302 not required)	PSV-302 is dropped only because nothing can isolate V-101.	Confirm on the P&ID and the isometrics that no block valve is fitted; otherwise add PSV-302 (fire + blocked outlet).
EQ-R06	E-102 steam-side rating vs letdown failure (disagree)	EQL MEA-ME-EQL-001 (E-102 tubes 420 kPa g, from the LP header pressure); SRS SIF-3 demand case (10 bar a MP steam); Safety layers (PSV-207 Gap); this RLB	With the EQL rating a letdown failure overpressures the steam side, and PSV-207 sized for FV-207 wide open on MP steam needs several T orifices (part 4).	Re-rate the E-102 tube side and the FV-207 downstream piping for the MP header in the EQL (then no PSV-207), or confirm a full-flow relief on the LP header at the letdown station that limits it to the EQL rating.
EQ-R07	P&ID note 5 lists a PSV on E-101	CC-PID-101 note 5; this RLB	E-101 is a plate exchanger with liquid on both sides: a plate leak is not an overpressure case. Only thermal expansion of the blocked-in rich side needs a relief device.	Show TSV-501 on the rich side; delete "E-101" from the PSV list of note 5.
EQ-R08	Steam valve pressure drop in the model	src/model.js (reboiler condenses at the header pressure); MEA-IN-INL-001 (FV-207 sizing)	Because FV-207 takes no pressure drop in the model, the reboiler pinch at relieving pressure is the most favourable it can be. A real valve ΔP lowers the condensing temperature and the blocked-outlet load.	Keep the model load (conservative). Recompute when the steam-side hydraulics are issued.
EQ-R09	Fire wetted area uses assumed elevations	This RLB (fire case); Layout & isometrics (not issued)	The fire load depends on how much of T-102 and V-101 lies within 7.6 m of grade.	Confirm the elevations in the Layout tab and recompute (the fire calculator in part 3 takes them as inputs).
EQ-R10	No CO2 vent on the P&ID	CC-PID-101; Safety layers (VENT-CO2); Consequence tab (CO2 dense plume)	The product line has no start-up / compressor-trip vent and the PSV discharge location is not shown.	Add HV-305 and the vent stack VS-301 (shared with PSV-301) at a high point; check dispersion in the Consequence tab.
EQ-R11	E-102 and E-103 tube rupture (agree: credible, not governing)	EQL MEA-ME-EQL-001 EQ-M01, EQ-M02; this RLB (part 3)	Both are credible by the 10/13 rule with the EQL design pressures. Steam at the normal LP pressure cannot enter a shell relieving at set + 10 % (zero load). Cooling water does enter E-103, but as liquid it only displaces vapour, so its relief load is small. Neither governs PSV-301.	Agreed with the EQL that they are relief cases: rows TR and TR2 of the PSV-301 case table. Confirm the CW supply pressure (the EQL assumes 400 kPa a).

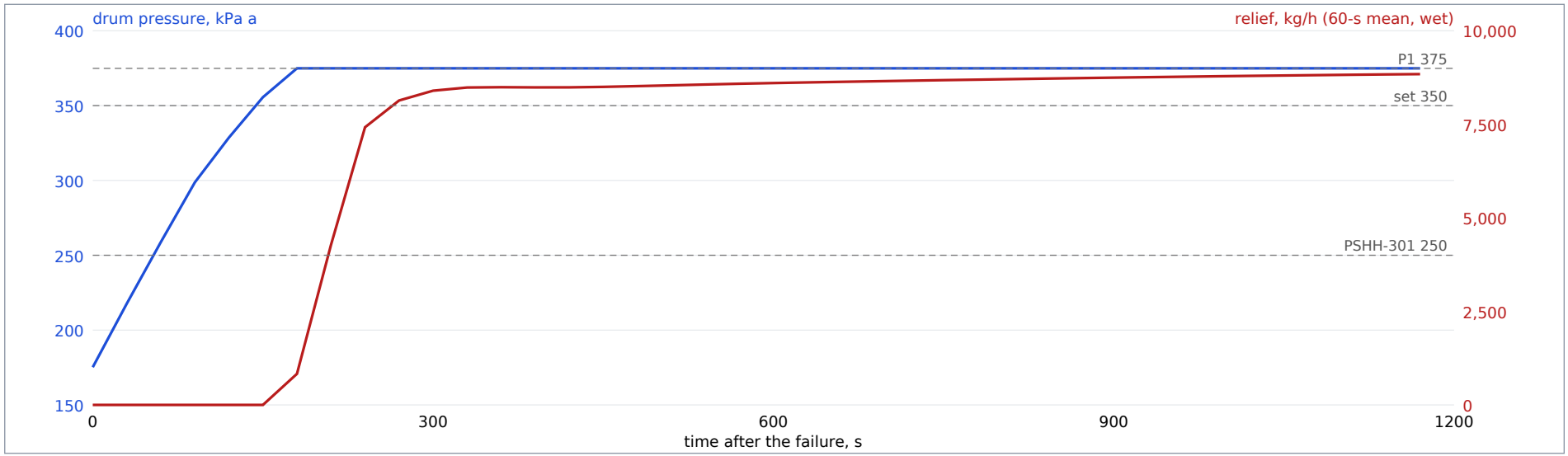
EQ	Subject	Documents that disagree	What	Suggested resolution
EQ-R12	E-104 tube rupture: amine into the CW side (partly disagree)	EQL MEA-ME-EQL-001 EQ-M02 (relief case for the CW return); this RLB	Credible by the 10/13 rule, but at its maximum operating pressure the lean side cannot exceed the CW-side MAWP, so no relief load results. Overpressure would need P-102 at shut-off with FV-201 closed as well (two failures).	No relief device for this case. Add amine detection on the CW return (TBC) and keep TSV-502 for thermal expansion only.

9 Assumptions (not design data)

Item	Value
MAWPs	From MEA-ME-EQL-001 except the E-102 steam-side MP re-rating (assumed): T-102, E-102 shell (process side), E-103 process side and V-101: 270 kPa g + FV (EQL MEA-ME-EQL-001; op + 170 kPa governs); E-101 rich side (TSV-501): 580 kPa g (EQL MEA-ME-EQL-001; P-101 shut-off (677 kPa a)); Cooling-water sides of E-103, E-104, E-105: 470 kPa g (EQL MEA-ME-EQL-001; CW supply 400 kPa a assumed there); E-102 tube side (LP steam), re-rated for the MP header: 10 bar g (assumed, proposed by this RLB); E-102 tube side (LP steam) as issued: 420 kPa g (EQL MEA-ME-EQL-001; LP header 350 kPa a); T-101 shell: 190 kPa g (EQL MEA-ME-EQL-001; op + 170 kPa)
Overpressure	10 % (single valve, non-fire), 21 % (fire), 10 % (thermal)
Coefficients	K_d 0.975 gas/steam, 0.65 liquid; K_b 1 (conventional, atmospheric discharge); K_c 1 (no rupture disc); Z 1
Gas properties	CO2 + water vapour, ideal gas, c_p 37.4 / 34.0 J/(mol K) at about 100 °C (typical values) for k
Fire	Fire zone 7.6 m above grade (API 521). Assumed: kettle/sump bottom at grade +1 m; T-102 packing starts 6.0 m above grade; V-101 horizontal, D 1.6 m × L 4.0 m (8.0 m³ = K.Vdrum), at grade; F = 1 (bare), adequate drainage and fire fighting. Liquid at the high alarm level (LIC-208 H, LIC-303 H).
Thermal expansion	α_v 5.0e-04 1/K (aqueous MEA), 3.5e-04 1/K, ρ 994, c 4180, μ 0.72 cP (cooling water); amine ρ, c, μ from the model
MP steam (letdown failure)	MP steam 10 bar a / 179.9 °C in the LP header (letdown failure): the demand case of SIF-3 in the SRS tab (assumed there).
Tube rupture	tube ID 15.75 mm, two open ends, K_d 0.62
Vent stack	ID 202.7 mm (DN 200), ambient 30 °C

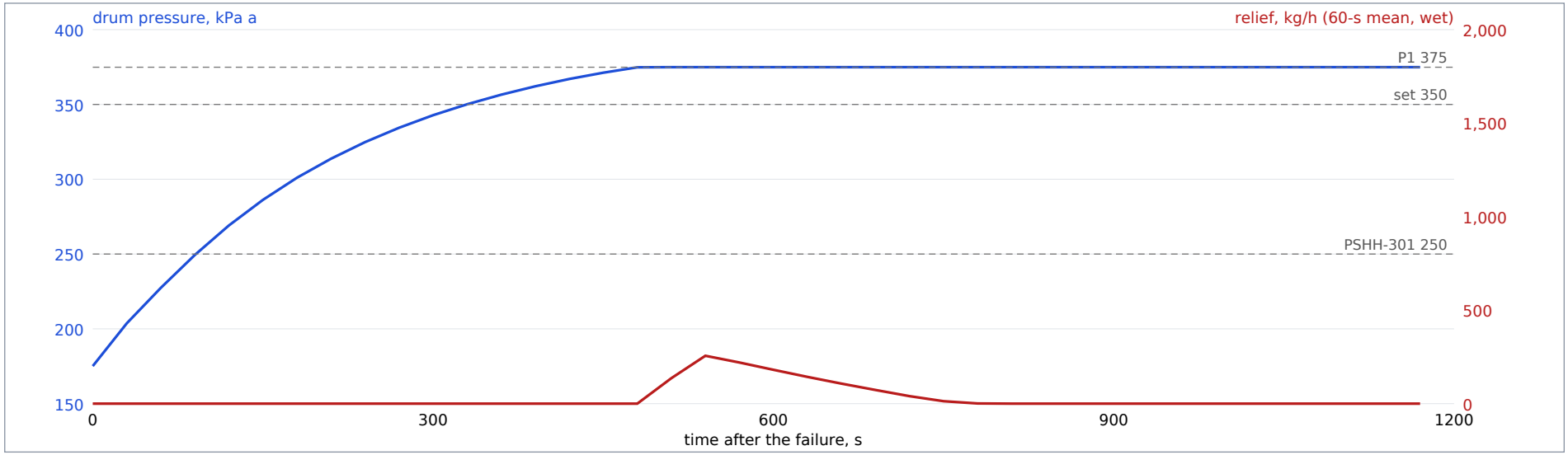
Appendix A · Model relief runs (traces)

R1 · Blocked outlet: PV-301 closed; FV-207 held at its normal opening



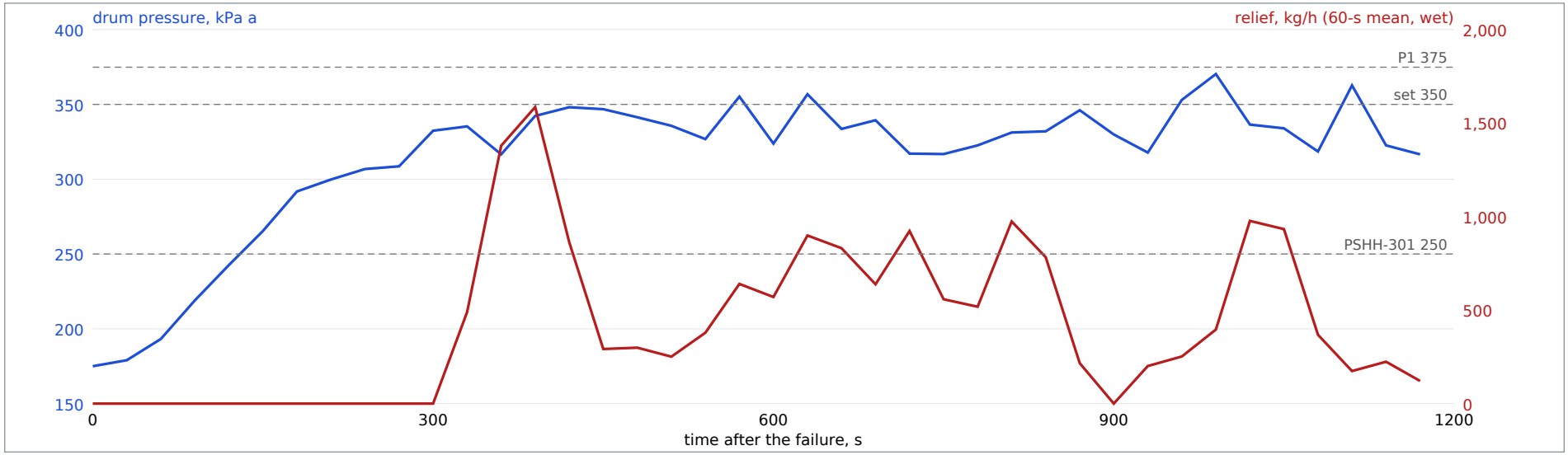
Lift at 170 s; largest 60-s mean relief 8,848 kg/h wet (7,147 dry). Run from the design state, trips disabled, logged every 30 s.

R3 · Blocked outlet after the I-2 trip: PV-301 closed and XV-207 shut at t = 0



Lift at 481 s; largest 60-s mean relief 260 kg/h wet (240 dry). Run from the design state, trips disabled, logged every 30 s.

R7 · General power failure: all motors and the CW pumps stop; steam stays on, PV-301 held



Lift at 316 s; largest 60-s mean relief 2,076 kg/h wet (1,693 dry). Run from the design state, trips disabled, logged every 30 s.